

ML Week 11

Number of participants: 27



1. In fitting of neural networks:

5 correct answers
out of 19 respondents



Back-propagation simply computes the gradient of the parameters of a neural network.



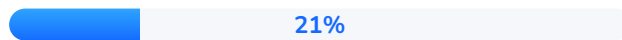
13 votes

The forward pass has to be done only once per training sample in the whole gradient descent.



5 votes

Stochastic gradient descent computes the exact gradient faster than in usual gradient descent.



4 votes



Back-propagation is efficient because many calculations can be stored and re-used.



15 votes



2. In regularization for neural networks:

8 correct answers
out of 16 respondents

Neural networks
have a convex loss
function.



13%

2 votes

Early stopping sets
some model
parameters to zero.



13%

2 votes



Dropout can be
used together with
L1 regularization.

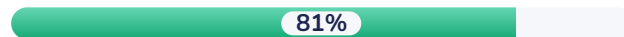


69%

11 votes



The initial
parameter values
have a strong
influence on the
model
performance.



81%

13 votes



3. Which the following statement(s) is/are correct concerning backpropagation?

9 correct answers
out of 19 respondents

It is an approximation of the gradient of the neural network.



11%

2 votes

It efficiently computes the gradient of the neural network using the chain rule for differentiation.



84%

16 votes

It can reuse quantities computed at an earlier step.



74%

14 votes



4. Which of the following statement(s) is/are correct?

14 correct answers
out of 19 respondents

The softmax activation function is usually used for regression tasks.



26%

5 votes

Dropout always increases the training performance of a neural network.



0%

0 votes

Neural networks are parametric models.



74%

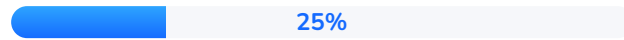
14 votes



5. Have you filled out the evaluation of the course?

20 respondents

Yes



25%

5 votes

No



75%

15 votes